

HOJUN • CLASS 12

# Offshore, Worldwide

Four wind farms, four seabeds. You be the engineer.

## WARM-UP

# Catching Up

## QUESTIONS

**Q1** How's the week been on site?

**Q2** Anything that turned out harder than planned?

## TOOLBOX

**It's been ~ so far**

지금까지는 ~해요

**~ took longer than we planned**

~가 계획보다 오래 걸렸어요

FROM YOUR WORDS · 지난주 단어

wise

universal

CAUGHT TODAY

# Hornsea 2

Ørsted • 2022 • 1.32 GW • 165 turbines at 8 MW • the world's largest when it opened

Monopiles 60–78 m long, depending on water depth • 390 km of cable • sits next to Hornsea 1 (1.2 GW)

Hornsea 3's monopiles: SeAH — a Korean company

**Q1** Sixty-metre monopiles versus seventy-eight — what changes when you go longer?

↳ Length is water depth plus how deep it's driven plus height above the water. Longer means heavier, a bigger hammer, a bigger vessel, and a narrower weather window.

**Q2** 165 turbines, 390 km of cable. Where's the biggest risk?

↳ In the cables. Most offshore wind failures are cable faults — anchors, trawl nets, bad burial. Turbines rarely break. Cables do.

**Q3** SeAH makes the monopiles for Hornsea 3. What does that tell you?

↳ Korean steel pipe is already inside the European supply chain. The next step is Korean installation and engineering — not just the steel.

## TOOLBOX

- The longer the pile, the ~
- At that distance, ~
- What changes at depth is ~
- The risk moves to ~
- You'd never get away with ~
- The clever part is ~
- I'd trust ~ before ~
- If I had to bet, ~

# Dogger Bank

**3.6 GW across phases A, B, C · 130–200 km offshore · water 18–63 m deep · Haliade-X 13–14 MW turbines**

Monopiles up to 72.8 m long and 1,424 tons · an 8-metre flange, the largest ever · 277 monopiles in total

Chosen because it's far enough to be invisible from land, yet shallow enough for fixed foundations

## Q1 130 km out. How does that change the work?

↳ A crew transfer takes hours each way, so you use service vessels that stay out for weeks. Every trip is expensive, so every mistake is expensive.

## Q2 A 1,424-ton monopile with an 8-metre flange. What can go wrong on the lift?

↳ Crane capacity, swell moving the vessel, keeping the pile plumb while driving — and damaging the flange, the one part that has to be perfect for the transition piece to bolt on.

## Q3 Why pick a site nobody can see?

↳ No visual complaints, stronger and steadier wind — about 10 m/s — and yet only 18 to 63 metres deep. Far away, but still standing on the seabed.

### TOOLBOX

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# Borssele 1+2

Ørsted • 2020 • 752 MW • 94 turbines at 8 MW • 23 km off the coast of Zeeland

Monopile foundations • 94 inter-array cables, each buried two metres into the seabed

Built with jack-up vessel Innovation and cable-layer Nexus

## Q1 Ninety-four cables buried two metres deep. Why bury them?

↳ Ship anchors and fishing gear are the number-one cause of cable damage. Two metres keeps them out of reach — and insurers won't cover an unburied cable.

## Q2 Only 23 km from shore. What's easier, and what's harder?

↳ Easier: short cables, quick transfers, cheaper maintenance. Harder: more shipping, more fishing, more people watching, tighter permits.

## Q3 94 turbines at 8 MW. Why not fewer, bigger ones?

↳ In 2020, 8 MW was the largest proven turbine. Today the same farm would use half as many 14–15 MW units — fewer foundations, fewer cables, fewer lifts.

### TOOLBOX

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# Hywind Scotland

**Equinor • 2017 • the world's first floating wind farm • 5 turbines at 6 MW • 30 MW • 29 km off Peterhead**

Water 95–120 m deep • spar-buoy foundations: 90 m of steel below the surface, 3,500 tons each, held by three mooring chains

Highest capacity factor of any UK offshore wind farm (54%) • cost £264m

**Q1** 100-metre water. What does a monopile do there?

↳ *Nothing — it can't. Fixed foundations stop making economic sense around 40 to 50 metres. Below that, you float.*

**Q2** A spar buoy — 90 m of steel underwater, three mooring chains, nothing in the seabed. Would you trust it?

↳ *It's oil-and-gas technology; the same idea has held North Sea platforms for decades. The weight sits low, so it doesn't tip. The risk moved from the seabed to the mooring lines.*

**Q3** Highest capacity factor in the UK. Why would floating beat fixed?

↳ *Deep water is far from shore, and far from shore means stronger, steadier wind. Floating lets you go where the wind actually is.*

## TOOLBOX

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## YOUR VERDICT

# Four Seabeds, One Engineer

### QUESTIONS

- Q1** Of the four, which one was the hardest to build?
- Q2** Which one would you want to work on?
- Q3** Which one would keep you up at night?
- Q4** In ten years, will most new wind farms stand on the seabed, or float?
- Q5** The sea off Korea's east coast gets deep very quickly. What does that mean for wind farms there?

↳ Little shallow water on the east coast — so the big projects off Ulsan are going floating.

### TOOLBOX

**I'd say ~ was the hardest, because ~**

제일 어려웠던 건 ~예요, 왜냐하면 ~

**I'd want to be on ~**

~에서 일해보고 싶어요

**~ would worry me — ~**

~는 걱정될 거예요 — ~

**Ten years from now, ~**

10년 뒤엔 ~

**For Korea, that means ~**

한국에선 그게 ~라는 얘기죠

### CAUGHT TODAY